

EvidITI: Enhancing RAG Robustness through Inference-Time Intervention on Evidence-Seeking Attention Heads

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Abstract. Retrieval-Augmented Generation (RAG) significantly extends the capabilities of Large Language Models (LLMs) by incorporating external knowledge. However, RAG is vulnerable to noisy retrieval, triggering "Hallucination on Hallucination" and degrading reliability. Existing solutions rely on costly fine-tuning or high-latency prompting. Inspired by Inference-Time Intervention techniques, this paper extends an ITI framework aimed at enhancing evidence adherence in RAG scenarios. By identifying specific attention heads responsible for evidence processing using a model-error-driven dataset, we apply linear shifts to their activations during inference, steering the model to prioritize retrieved evidence over internal hallucinations. Extensive experiments on Natural Questions, TriviaQA, and PopQA show our method improves robustness under up to 80% noise with negligible latency (<4%) and strong cross-dataset generalization, without modifying model parameters.

Keywords: Retrieval-Augmented Generation · Inference-Time Intervention · Robustness · Evidence Adherence · Hallucination Mitigation

1 Introduction

While Large Language Models (LLMs) excel in various tasks, their reliance on parametric knowledge often leads to hallucinations and obsolescence [1]. Retrieval-Augmented Generation (RAG) mitigates this by embedding external evidence into the context [16]. However, RAG robustness remains a bottleneck; retrieval noise can trigger "Hallucination on Hallucination" [4, 6], where misleading information amplifies a model's inherent errors (see Fig. 1).

Existing robustness solutions face core trade-offs: supervised fine-tuning (SFT) is resource-heavy and harms generalization, while prompt-based methods (e.g., Chain-of-Note [15]) add latency and cannot override the model's stubborn internal priors. As LLMs' internal activations hold latent truth-discriminating structures [11], directly intervening on these representations can enhance evidence adherence without parameter updates.

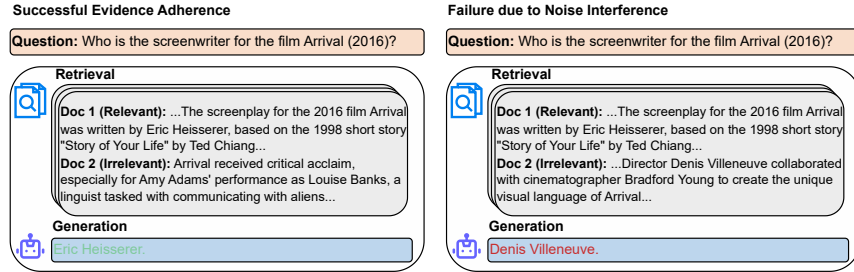


Fig. 1. Comparison between successful evidence adherence and noise-induced failure in RAG.

Inspired by Inference-Time Intervention (ITI) techniques [9], we propose a geometry-based ITI framework for RAG scenarios. We identify evidence-processing attention heads via a Model-Error-Driven Dataset, then apply linear activation shifts during inference to steer the model toward evidence-constrained generation. Experiments on three benchmarks show our method improves noisy-environment robustness with negligible latency and strong cross-dataset generalization. The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 details the EvidITI framework; Section 4 presents experimental results and analysis; and Section 5 concludes the paper.

2 Related Work

RAG systems are vulnerable to diverse retrieval noise types [2]. Existing mitigation strategies fall into two main classes: SFT-based methods [14, 15] that incur high training costs, and training-free frameworks that add inference latency. While Knowledge Graph integration [13] and attention modification [3] show promise, they rely on external modules or heavy computation. Our method bypasses these limitations via internal activation editing. Our work builds on Inference-Time Intervention (ITI) [9], a minimally invasive technique that aligns model generation with its internal factual knowledge via activation shifts. We extend ITI from general factual alignment to evidence adherence in RAG scenarios, enabling precise control over the model’s evidence utilization.

While recent methods like Self-RAG and CrAM [3] improve robustness through fine-tuning or specialized attention masks, EvidITI distinguishes itself by being entirely training-free and model-agnostic, intervening only on internal activations without altering the model’s architecture or requiring high-performance decoders.

3 Methodology

This section details an Inference-Time Intervention framework designed to enhance the "Evidence Adherence" capability of Large Language Models in Retrieval-Augmented Generation scenarios. The overall architecture of our method is illustrated in Fig. 2. Our method does not alter model parameters; instead, during the inference stage, it applies linear shifts to the activation values of specific attention heads, guiding the model to focus more on retrieved evidence rather than its internal hallucinations or incorrect knowledge.

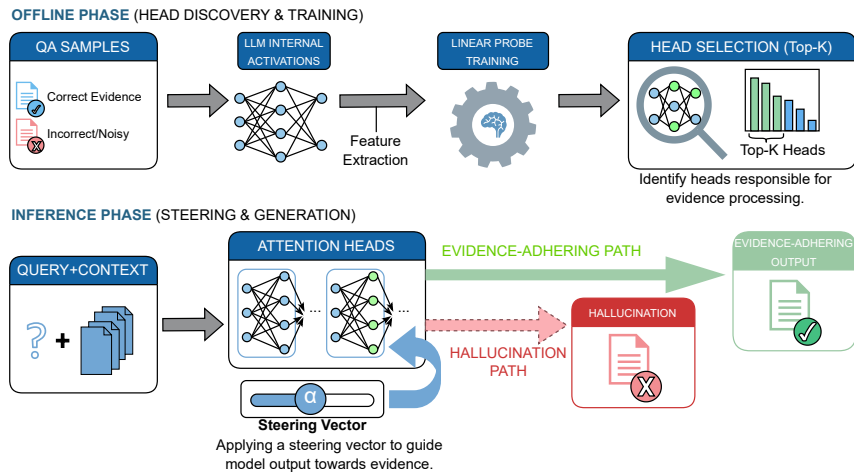


Fig. 2. The proposed Inference-Time Intervention framework for RAG. The process is divided into: (1) **Offline Phase**, where we extract internal activations and train linear probes to identify Top-K attention heads responsible for evidence adherence; and (2) **Inference Phase**, where a steering vector is applied to the selected heads to guide the LLM toward generating evidence-based responses.

3.1 Dataset Construction and Robustness Test Benchmark

To train probes capable of distinguishing between "evidence-adhering" and "evidence-ignoring" behaviors, and to evaluate the model's robustness in noisy environments, we constructed specialized training and test sets.

Training Set Construction. We construct a Model-Error-Driven Dataset by leveraging the performance gap between the target LLM and high-performance validation models (e.g., GLM-4). For a given query and retrieved documents, responses from the target LLM are validated; incorrect ones are labeled as negative

samples (hallucinations), while correct ground-truth answers are labeled as positive. Since PopQA focuses on long-tail knowledge, correct answers are primarily attributed to successful evidence utilization rather than parametric memory, making this dataset an effective proxy for evidence adherence. Regarding probe data, although the validation model may have inherent biases, using it as a 'silver standard' effectively captures the relative gap between noise-sensitive and evidence-adhering behaviors.

Robustness Test Set Construction. To evaluate the effectiveness of our method under retrieval noise, we constructed test sets with varying proportions of noise. For each query, we control the number of relevant documents and irrelevant noisy documents in the retrieval context. The test set includes different noise levels from $N = 0$ to $N = 5$, where N represents the number of irrelevant documents inserted.

3.2 Linear Probe Training and Head Selection

To achieve precise intervention, we need to identify the key attention heads responsible for processing evidence. This process corresponds to the **Offline Phase** in Fig. 2.

Activation Collection and Feature Extraction. We first convert raw model behaviors into a structured feature space. Given an input sequence x consisting of a query and retrieved context, we feed both positive samples (evidence-adhering) and negative samples (hallucinated/noisy) into the LLM. For the l -th layer and h -th attention head, we extract its output activation vector at the last prompt token as the representative feature $z_{l,h} \in \mathbb{R}^d$. This results in a feature dataset $\mathcal{D}_{act} = \{(z_i, y_i)\}_{i=1}^M$, where $y \in \{0, 1\}$ represents the model's adherence to evidence.

Probe Training. In the "Linear Probe Training" stage, we train an independent linear classifier $P_{l,h}$ for each head. The probe seeks to find a hyperplane in the extracted feature space that separates truth-telling states from hallucination states. We optimize the parameters $\theta_{l,h}$ by minimizing the cross-entropy loss:

$$\mathcal{L}(\theta_{l,h}) = - \sum_{i=1}^M [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where $\hat{y}_i = \text{sigmoid}(\theta_{l,h}^T z_i)$.

Head Selection Strategy. As shown in the "Head Selection (Top-K)" module, we identify the heads most "responsible" for evidence processing. We evaluate each probe's accuracy $Acc_{l,h}$ on a held-out validation set. Heads are ranked by $Acc_{l,h}$, and the top K heads are selected as the intervention target set $\mathcal{H}_{topK}$. This ensures our intervention is localized only to the most informative components of the model.

3.3 Inference-Time Intervention Mechanism

During the inference stage, we apply intervention on the selected attention head set $\mathcal{H}_{topK}$ to shift the model’s internal state towards "evidence adherence."

Intervention Direction Calculation. In high-dimensional spaces, the class geometric center better represents feature characteristics than the probe normal vector. We thus use the Center-of-Mass mean shift as the intervention direction $v_{l,h}$, defined as the difference between the mean activations of positive and negative samples:

$$v_{l,h} = \frac{\mu_{pos} - \mu_{neg}}{\|\mu_{pos} - \mu_{neg}\|_2} \quad (2)$$

where $\mu_{pos} = \mathbb{E}[z_{l,h}|y = 1]$ and $\mu_{neg} = \mathbb{E}[z_{l,h}|y = 0]$.

Activation Editing. During the autoregressive generation process, for each layer l and head h , if $(l, h) \in \mathcal{H}_{topK}$, we superimpose an intervention vector onto its original activation output $z_{l,h}$:

$$\tilde{z}_{l,h} = z_{l,h} + \alpha \cdot \sigma_{l,h} \cdot v_{l,h} \quad (3)$$

where:

- α is a hyperparameter controlling the intervention strength. α controls intervention strength: larger values enforce stricter evidence adherence, while smaller values retain more of the model’s original behavior.
- $\sigma_{l,h}$ is the standard deviation of the activation projection for that head on the validation set, used to normalize the intervention scale.

Hyperparameter Search. The intervention strength α and the number of intervened heads K are optimized via grid search on the validation set to maximize EM and F1 scores. Notably, optimal K and α are determined solely on the PopQA validation set and then fixed for all other evaluation benchmarks to demonstrate true cross-dataset generalization.

4 Experiments

4.1 Experimental Setup

Datasets and Metrics. We evaluate our method on three widely used open-domain question answering (ODQA) benchmarks: Natural Questions (NQ) [8], TriviaQA [7], and PopQA [10]. To verify the generalization ability of our probes, we use PopQA to construct the training set for learning the intervention heads and directions, and evaluate on all three datasets. We adopt Exact Match (EM) and F1 score as evaluation metrics, which are widely used in the QA setting.

Noise Construction. To simulate real-world retrieval scenarios where the retriever may return irrelevant or noisy documents, we construct robustness test sets with varying noise levels. For each query, we retrieve a fixed number of documents but control the ratio of irrelevant passages. We test noise ratios ranging from 0% (all documents are relevant) to 100% (all documents are irrelevant), with 80% noise being the default setting for our main results unless otherwise stated.

We test noise ratios from 0% to 100%, with 80% noise as the default for main results, consistent with the setup in Section 3.1.

Baselines and Models. We compare against three baselines: Naive LLM, standard Naive RAG, and Chain-of-Note (CoN) [15]. Experiments are conducted on Llama-2-7b-chat [12], Llama-3-8B-Instruct [5], and Vicuna-7b-v1.5 [17].

4.2 Main Results

Main evaluation uses 80% noise to simulate challenging retrieval environments; 100% noise is excluded from Table 1 (full trends analyzed in Section 4.3).

Performance Improvement. Our method consistently outperforms the baselines in most settings.

- On **Llama-2-7b-chat**, our method achieves the best performance across all datasets, outperforming Naive RAG by 1.63% F1 and CoN by 0.41% F1 on PopQA.
- On **Vicuna-7b-v1.5**, improvements are most significant: we outperform Naive RAG by **6.99%** F1 on PopQA and **4.47%** F1 on TriviaQA, showing our method is especially effective for models with weaker evidence-utilization capabilities.
- On **Llama-3-8B-Instruct**, our method remains competitive, outperforming baselines on TriviaQA (+0.47% F1) and PopQA (+0.83% F1). Smaller gains on NQ stem from Llama-3’s strong inherent instruction-following capabilities, which leave less intervention space.

These results confirm that intervening on evidence-seeking heads improves the model’s context utilization across diverse model architectures.

4.3 Robustness to Retrieval Noise

We evaluated Llama-2-7b-chat across 0%-100% noise ratios; Fig. 3 shows our method’s stronger robustness over Naive RAG.

- **Consistency:** Our method gains +1.59% F1 on NQ at 0% noise, proving intervention does not harm clean evidence processing.
- **Resilience:** The performance gap widens with higher noise, with a +1.38% F1 lead on NQ at 40% noise.

Table 1. Performance comparison (EM/F1) across NQ, TriviaQA, and PopQA datasets under a high-noise retrieval environment (80% noise ratio). The best results for each model backbone are highlighted in **bold**.

Model	Method	NQ (%)		TriviaQA (%)		PopQA (%)	
		EM	F1	EM	F1	EM	F1
Llama-2-7b-chat (<i>top-k=35, $\alpha=7$</i>)	Naive LLM	11.33	22.51	46.33	53.51	19.67	24.91
	Naive RAG	39.67	54.76	68.00	74.56	63.67	69.89
	CoN	38.33	53.41	54.67	66.89	65.00	71.11
	Ours	42.33	56.19	68.33	75.08	66.00	71.52
Llama-3-8B-Instruct (<i>top-k=60, $\alpha=3$</i>)	Naive LLM	18.33	28.26	64.00	68.66	25.00	30.03
	Naive RAG	51.33	66.39	77.67	83.81	63.67	72.08
	CoN	53.33	67.36	73.00	78.78	60.67	68.46
	Ours	50.67	65.06	78.33	84.28	64.66	72.91
Vicuna-7b-v1.5 (<i>top-k=120, $\alpha=7$</i>)	Naive LLM	6.00	16.34	26.33	40.42	3.67	12.32
	Naive RAG	10.00	29.07	43.00	55.91	37.33	52.69
	CoN	13.33	30.58	39.00	52.39	38.00	50.90
	Ours	20.33	34.98	49.67	60.38	50.00	59.68

- **Extreme Cases:** At 100% noise, our method does not amplify hallucinations, confirming it reinforces evidence-following rather than blind context adherence.

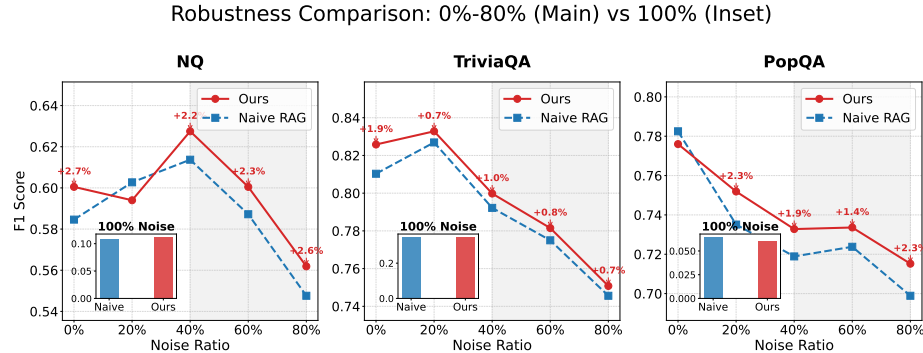


Fig. 3. Robustness analysis of Llama-2-7b-chat across varying noise levels (0% to 100%). Our ITI-based method exhibits superior resilience compared to the Naive RAG baseline as the proportion of irrelevant context increases.

4.4 Ablation Study

Ablation studies on Llama-2-7b-chat (PopQA, 80% noise) validate our design choices; results are summarized in Table 2.

- **Direction Matters:** Random or reversed intervention directions cause 1.58% and 2.30% F1 drops, confirming our probe-learned direction encodes evidence adherence semantics.
- **Localization Matters:** Randomly selected heads cause catastrophic performance degradation, proving only a sparse subset of heads controls evidence processing.
- **Accuracy-based Selection:** Bottom-K heads underperform our Top-K strategy, validating probe accuracy as a reliable indicator for critical head selection.

Table 2. Ablation study on the PopQA dataset (80% noise) using Llama-2-7b-chat. We compare our proposed Top-K selection and Center-of-Mass (CoM) direction against various alternative configurations.

Variant	EM (%)	F1 (%)
Ours (Top-K Heads + CoM Dir)	66.00	71.52
Reverse Direction	62.33	69.22
Random Direction	63.67	69.94
Random Heads	6.33	16.29
Bottom-K Heads	63.67	70.20

4.5 Hyperparameter Sensitivity

We analyzed the impact of intervention head count K and strength α ; Fig. 4 shows F1 scores on PopQA, with a clear optimal "sweet spot" for these hyperparameters. Although we do not explicitly report Perplexity, the stable performance within this range suggests that the model’s fundamental language modeling capability remains intact. For Llama-2-7b-chat:

- **Head Count (K):** Performance peaks at $K=32-48$; excessive K (>64) degrades performance by disrupting general language modeling.
- **Strength (α):** Moderate values ($\alpha \in [3, 7]$) are optimal; excessive strength combined with large K causes significant performance drops.

4.6 Inference Efficiency

Our method only involves lightweight matrix additions during the forward pass, with a negligible 3.22% latency overhead over Naive RAG (Table 3), making it suitable for real-time applications.

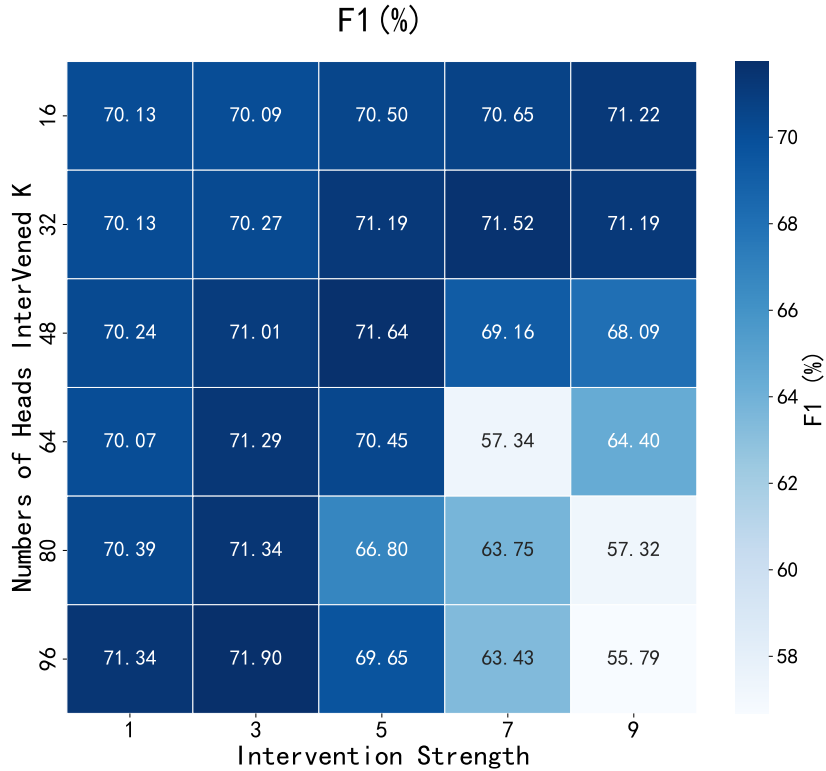


Fig. 4. Sensitivity analysis of hyperparameters K (number of intervened heads) and α (intervention strength) on the PopQA dataset. The heatmap indicates an optimal "sweet spot" for steering the model without collapsing its generative capabilities.

Table 3. Inference latency evaluation on 300 samples from PopQA. The additional computational overhead introduced by our intervention mechanism is negligible ($<4\%$) compared to the standard RAG pipeline.

Method	Inference Time (300 samples)	Additional Overhead
Naive RAG	127.00s	—
Ours	131.09s	+3.22%

5 Conclusion

We propose a geometry-based Inference-Time Intervention mechanism to enhance RAG noise robustness. By steering evidence-seeking attention heads, our method prioritizes retrieved context over hallucinations with negligible latency ($< 4\%$) and strong cross-dataset transferability. Future work includes refining probe precision, mitigating counterfactual retrieval risks, and developing adaptive intervention mechanisms.

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